Modified coherent synchrotron radiation point-kick model with self-consistent consideration of bunch length varying

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Coherent synchrotron radiation (CSR) in dipoles can degrade beam quality by increasing energy spread and transverse emittance. A simple one-dimensional CSR-kick model was developed to analyze CSR effects in dipoles and derive conditions for CSR suppression [Jiao *et al.* Phys. Rev. ST Accel. Beams **17**, 060701 (2014)]. However, this model assumes a constant bunch length within the dipole, limiting its accuracy in scenarios involving bunch compression, where the bunch length typically varies. This limitation requires numerical corrections, which may reduce the general applicability of the model. We present a modified CSR-kick model that accounts for bunch length variations within dipoles. The modified model is validated through simulations and consistently reproduces CSR suppression conditions for double-bend achromat (DBA) compressors, showing excellent agreement with previously numerically corrected results. Further analysis of the DBA compressor under CSR suppression conditions is also provided.

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I. INTRODUCTION

Coherent synchrotron radiation (CSR) imposes significant constraints on maintaining beam quality during the transport of high-brightness electron bunches in the magnetic deflection systems. Typically, when an electron bunch travels along a curved path, CSR emitted by the trailing particles can interact with the leading particles within the dipole. If not well controlled, CSR in bending magnets can degrade beam quality and lead to microbunching instability [1–9], increased energy spread, and transverse emittance growth [10–16].

Notably, CSR-induced projected emittance growth can be suppressed or eliminated under specific conditions known as CSR-kick cancellation conditions. Based on early insights into one-dimensional steady-state CSR, Douglas [17] proposed the concept of cell-to-cell phase matching to compensate for or cancel CSR kicks, which was further developed using the Courant-Snyder formalism [18]. Moreover, this emittance growth can also be compensated in a single achromatic cell by matching the beam envelope to the tilt of the transverse phase-space ellipse caused by CSR [19]. Building on these studies, an investigation [20] proved that beam envelope matching and Courant-Snyder analysis are equivalent and indicated that the complex CSR effects in a dipole can be simplified as a point kick. This point-kick model has been successfully applied to many scenarios to suppress steady-state CSR-induced emittance growth [21–26]. This model works well for CSR calculations in transport systems where the bunch length remains constant or changes very little, such as in double-bend achromats (DBAs) and triple-bend achromats [21,22].

However, it is not self-consistent in systems with a significant bunch length variation, particularly in magnetic bunch compressors. Therefore, this model is, in principle, no longer applicable to such systems. Nonetheless, one way to apply this point-kick analysis to these systems is to disregard self-consistency and assign different effective bunch lengths to dipoles within the system. Building on this idea, studies have shown that this model can be used to derive explicit CSR-kick cancellation conditions for bunch compressors [23–27]. When the bunch length varies significantly, these conditions will deviate from reality, and further numerical integrations will be necessary to make corrections [24]. While this correction method is effective, it is essentially a compromise. Moreover, the applicability of the numerically corrected results may be limited. Since the analysis is based

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on the nonself-consistent model, the generalizability of these numerical correction results is uncertain.

To address this issue, we propose a self-consistent modified CSR point-kick model that accounts for variations in bunch length. By calculating the bunch length as a beam with an energy chirp passes through a dipole magnet, we derive specific expressions for the horizontal coordinate and energy deviation $(x_{\mathrm{csr}}, x'_{\mathrm{csr}}, \delta_{\mathrm{csr}})$ under the influence of 1D steady-state CSR, and then the correction functions for the CSR point-kick model. With the modified analysis, the CSR-induced total kick and the corresponding emittance growth in a magnetic deflection system, especially during a bunch compression process, can be accurately predicted. We compared the modified model with simulation results and found that the theoretical predictions agree well with the simulations. Subsequently, we used this model to rederive the CSR-kick cancellation conditions for the DBA compressor, and the results are consistent with the numerically corrected results provided in Ref. [24].

Although the numerical fit coefficients in Ref. [24] are limited to smaller compression factors, our modified model demonstrates that they are also applicable to larger compression factors. Further analysis reveals that, under CSR-kick cancellation conditions, the first dipole contributes the majority of the bunch compression in a DBA compressor. For a given compression factor, there exists an upper limit to this contribution, beyond which the CSR-kick cancellation conditions can no longer be satisfied.

The remainder of this paper is organized as follows. Section II details the modified CSR point-kick model with varied bunch length taken into account. In Sec. III, we present the simulation validation of this model, demonstrating its improved suitability for bunch compressors and other magnetic systems with bunch length variations. Section IV discusses the CSR-kick cancellation conditions for the DBA compressor based on this model. Finally, we summarize our findings and offer concluding remarks in Sec. V.

II. MODIFIED CSR POINT-KICK MODEL

After a particle passes through a dipole magnet with a bending radius ρ and bending angle θ, its horizontal coordinate deviation $(x_{\mathrm{csr}}, x'_{\mathrm{csr}})$ and energy deviation δ_{csr} relative to synchronous particle change under the influence of CSR as follows [28–30]:

$$\begin{aligned} x_{\mathrm{csr}} &= \int_0^\theta \frac{d\delta_{\mathrm{csr}}(\phi)}{d\phi} R_{16}^{s\to s_f} d\phi, \\ x'_{\mathrm{csr}} &= \int_0^\theta \frac{d\delta_{\mathrm{csr}}(\phi)}{d\phi} R_{26}^{s\to s_f} d\phi, \\ \delta_{\mathrm{csr}} &= \int_0^\theta \frac{d\delta_{\mathrm{csr}}(\phi)}{d\phi} d\phi. \end{aligned} \tag{1}$$

Here, $R_{16}^{s\to s_f}$ and $R_{26}^{s\to s_f}$ are the elements of the dipole transfer matrices, representing the dispersion and dispersion gradient, respectively. The superscript $s \to s_f$ indicates the propagation from the current position s to the dipole exit s_f, with the corresponding angle variable changing from ϕ to θ. The gradient of the energy deviation due to CSR along the bending angle can be derived based on the 1D CSR model. When considering only the steady-state case and assuming the beam distribution in the dipole is Gaussian, it can be expressed as [31]

$$\frac{d\delta_{\mathrm{csr}}}{d\phi} = k\rho^{1/3}\left(\frac{\sigma_{z0}}{\sigma_z(\phi)}\right)^{4/3}. \tag{2}$$

Here σ_{z0} is the bunch length at the entrance of the dipole, $\sigma_z(\phi)$ denotes the bunch length after bending by an angle ϕ, and the CSR strength parameter k corresponds to the initial bunch length σ_{z0} and is in the form [31]

$$k = 0.2459\frac{N_b r_e}{\gamma\sigma_{z0}^{4/3}}, \tag{3}$$

where N_b is the electron population, r_e is the classical electron radius, and γ is the relativistic Lorentz factor. Considering the beam has an initial energy chirp h, under linear transport, the bunch length in Eq. (2) can be expressed as [32]

$$\begin{aligned} \sigma_z(\phi) &= \sqrt{[1+hR_{56}(\phi)]^2\sigma_{z_0}^2 + [R_{56}(\phi)\sigma_{\delta_0}]^2} \\ &\approx [1+hR_{56}(\phi)]\sigma_{z_0}. \end{aligned} \tag{4}$$

At the dipole exit, the bunch length after compression σ_{z_f} is related to σ_{z_0} with a compression factor $C \equiv \sigma_{z0}/\sigma_{zf} = 1/(1+hR_{56}^{s_0\to s_f})$. This approximation works when the beam is "cold" with low emittance, meaning the initial uncorrelated energy spread σ_{δ_0} is much smaller than the correlated energy spread $h\sigma_{z_0}$. When compression meets this condition, we call it linear compression. In Eq. (4), $R_{56}(\phi)$ represents the longitudinal dispersion from the dipole entrance to the current position $\phi(s)$ and can be expressed as

$$\begin{aligned} R_{56}(\phi) &= \int_0^\phi \frac{\eta(s)}{\rho} ds \\ &= \int_0^\phi \eta(\tau) d\tau \\ &= \int_0^\phi (\rho(1-\cos\tau) + \eta_0\cos\tau + \eta'_0\rho\sin\tau) d\tau \\ &\simeq \frac{\rho}{6}\left(\phi^3 + 6\frac{\eta_0}{\rho}\phi + 3\eta'_0\phi^2\right). \end{aligned} \tag{5}$$

Here, η_0 and η_0' are the dispersion function and its gradient at the entrance of the dipole, respectively. In the above expression, we assume that the dipole bending angle is small ($\theta \ll 1$), which holds true in most cases.

From the above analysis, by combining Eqs. (2)–(5), Eq. (1) can be rewritten as

$$\begin{pmatrix} x_{\mathrm{csr}} \\ x'_{\mathrm{csr}} \\ \delta_{\mathrm{csr}} \end{pmatrix} = k \begin{pmatrix} g_x \rho^{4/3}(\theta - \sin\theta) \\ g_{xp}\rho^{1/3}(1-\cos\theta) \\ g_\delta \rho^{1/3}\theta \end{pmatrix}. \tag{6}$$

Here, g_x, g_{xp}, and g_δ are the dimensionless correction functions, which can be obtained (for details, see Appendix),

$$\begin{aligned} g_x([a,b],\xi) &= 3\int_0^1 \left[1+\xi\frac{a\tau+b\tau^2+\tau^3}{a+b+1}\right]^{-4/3}(1-\tau)^2 d\tau, \\ g_{xp}([a,b],\xi) &= 2\int_0^1 \left[1+\xi\frac{a\tau+b\tau^2+\tau^3}{a+b+1}\right]^{-4/3}(1-\tau) d\tau, \\ g_\delta([a,b],\xi) &= \int_0^1 \left[1+\xi\frac{a\tau+b\tau^2+\tau^3}{a+b+1}\right]^{-4/3} d\tau, \end{aligned} \tag{7}$$

where the variable $\xi \equiv \sigma_{zf}/\sigma_{z0} - 1$ represents the rate of bunch length variation after passing through the dipole, the parameters a and b are related to η_0 and η_0', respectively, and can be written as

$$a = 6\frac{\eta_0}{\rho\theta^2}, \qquad b = 3\frac{\eta_0'}{\theta}. \tag{8}$$

They reflect the impact of the dipole on the dispersion function.

Considering the energy chirp of a beam at the dipole entrance is h, we have $\xi = h\rho\theta^3(a+b+1)/6$. Note that when the bunch length remains unchanged, i.e., $\xi = 0$, the correction functions (g_x, g_{xp}, g_δ) equal (1, 1, 1), and the modified model naturally degenerate to the constant-σ_z model in Ref. [20] as expected,

$$\begin{pmatrix} x_{\mathrm{csr}} \\ x'_{\mathrm{csr}} \\ \delta_{\mathrm{csr}} \end{pmatrix} = k \begin{pmatrix} \rho^{4/3}(\theta - \sin\theta) \\ \rho^{1/3}(1-\cos\theta) \\ \rho^{1/3}\theta \end{pmatrix}. \tag{9}$$

In this study, we mainly focus on theoretical discussions of the CSR effect in compression process, as sketched in Fig. 1.

Due to the arbitrariness of the parameters a and b in Eq. (8), the correction functions in Eq. (7) generally cannot be expressed in a simple analytical form. However, there are two special cases where the dispersion functions can be well determined to describe Eqs. (6) and (7). We first consider a dipole with zero entrance dispersion and dispersion gradient (e.g., the dipole in a single dipole system and the first dipole in a multidipole achromat system), where $a = b = 0$. In this case, the correction functions can be expressed in the following analytical form:

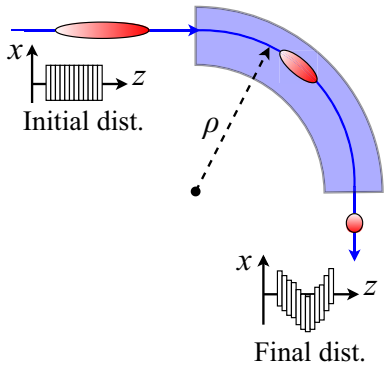


FIG. 1. Schematic of the compression process after a bunch passes through a dipole, with insets of the initial and final beam distribution affected by CSR into the $x-z$ phase plane (not to scale).

$$\begin{aligned} g_x([0,0],\xi) &= 3\frac{-1+(1+\xi)^{1/3}-\xi[{}_2F_1(\frac{1}{3},1,\frac{5}{3},-\xi)-1]}{\xi(1+\xi)^{1/3}}, \\ g_{xp}([0,0],\xi) &= \frac{2-{}_2F_1(\frac{1}{3},1,\frac{5}{3},-\xi)}{(1+\xi)^{1/3}}, \\ g_\delta([0,0],\xi) &\approx \frac{1}{(1+\xi)^{1/3}}, \end{aligned} \tag{10}$$

where ${}_2F_1$ denotes the standard hypergeometric function.

The first two terms in Eq. (10) are obtained through direct integration of Eq. (7). Although an analytical result for the g_δ function can also be derived, the expression is too complex and can be approximated in the last term of Eq. (10). The approximate expression is sufficiently accurate over a large range of ξ (see Fig. 2). Within the ξ range shown in Fig. 2, the deviation between the approximate expression and the integrated result is less than 0.3%. In addition to g_δ, we also provide g_x and g_{xp} for different ξ. We find that when ξ approaches -1 (corresponding to an infinitely large compression factor), all correction functions are greater than 1 and increase. Conversely, when $\xi > 0$ (corresponding to decompression process), all correction functions are less than 1 and decrease as ξ increases. This leads to the obvious conclusion that when the bunch length is reduced, the constant-σ_z model underestimates the CSR effect. Conversely, when the bunch length is increased, the constant-σ_z model tends to overestimate the CSR effect.

Another special case is when both the dispersion and dispersion gradient at the dipole exit are zero (e.g., the last dipole in a multidipole achromat system). In this case, we have $\eta_0 + \eta_0'\rho\theta + \rho\theta^2/2 = 0$ and $\eta_0' + \theta = 0$, where η_0 and η_0' are the dispersion functions at the entrance of the last

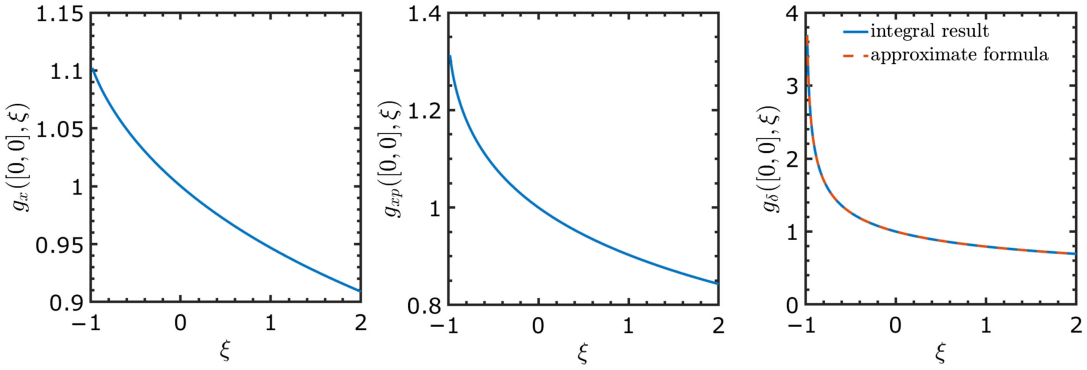


FIG. 2. The correction functions g_x, g_{xp}, and g_δ for the case of $a = b = 0$ as a function of the bunch length variation ξ within the range $\xi \in (-1, 2]$ (compression factor $C \in [\frac{1}{3}, \infty)$). The comparison between the approximate expression and the integrated result for g_δ is also shown.

dipole. Form Eq. (8), we obtain $a = 3$ and $b = -3$. The corresponding correction functions can be expressed analytically as

$$g_x([3,-3],\xi) = 3\frac{(1+\xi)^{1/3}-1}{\xi(1+\xi)^{1/3}},$$
$$g_{xp}([3,-3],\xi) \approx \frac{1}{(1+\xi)^{3/4}},$$
$$g_\delta([3,-3],\xi) \approx \frac{1}{1+\xi}. \tag{11}$$

For the g_{xp} and g_δ functions, we utilized approximate forms to avoid overly complex expressions. These approximations agree well with the integrated results (see Fig. 3), with the deviation for g_{xp} being less than 8% and for g_δ less than 0.02%.

The trend of the correction functions with ξ is similar with the previous case in Fig. 2, but the variation from 1 is larger for the same nonzero ξ. This indicates that the CSR effect is more sensitive to bunch length variation compared to the previous case.

To derive the CSR point-kick model, we follow the steps in Ref. [20], treating the CSR kick as an equivalent effect applied at the midpoint of the dipole.

Notably, Eq. (6) represents the result of CSR acting on the particle after it passes through the dipole. When considering the initial energy deviation of a particle as δ_0, its horizontal coordinate deviation will become

$$X_e = \begin{pmatrix} x_e \\ x_e' \end{pmatrix} = \begin{pmatrix} \rho(1-\cos\theta) \\ \sin\theta \end{pmatrix}\delta + \begin{pmatrix} g_x\rho^{4/3}(\theta-\sin\theta) \\ g_{xp}\rho^{1/3}(1-\cos\theta) \end{pmatrix}k, \tag{12}$$

where $\delta = \delta_0$ represents the energy deviation of the particle before being affected by this dipole and CSR.

To obtain the horizontal coordinate deviation after the CSR kick at the center of the dipole, $X_k = (x_k, x_k')$, we simply degenerate X_e by half of the dipole, i.e., $X_k = M_h^{-1}X_e$, where M_h^{-1} is the inverse transfer matrix for half of the dipole in (x, x') plane. In this way, we obtain

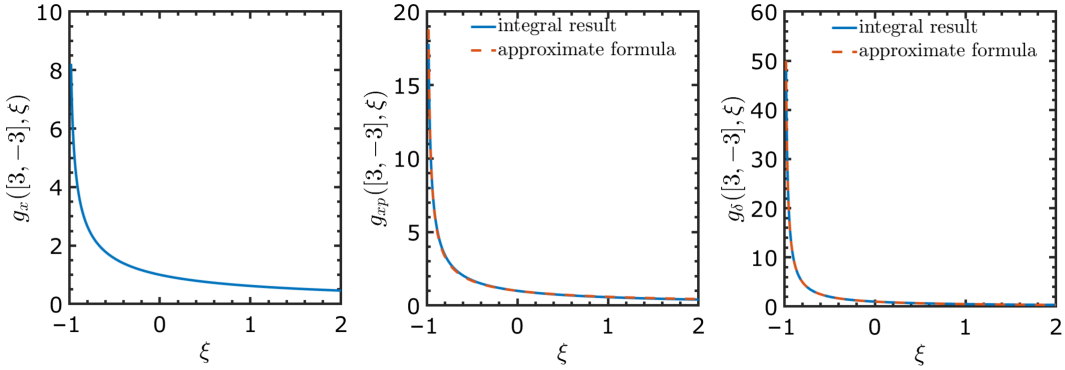


FIG. 3. The correction functions g_x, g_{xp}, and g_δ for the case of $a = 3$ and $b = -3$ as a function of the bunch length variation ξ within the range $\xi \in (-1, 2]$ (compression factor $C \in [\frac{1}{3}, \infty)$). The comparison between the approximate expressions and the integrated results for g_{xp} and g_δ are also shown.

$$X_k = \begin{pmatrix} x_k \\ x_k' \end{pmatrix} = \begin{pmatrix} \rho^{4/3} k[g_x \theta \cos(\theta/2) - 2(g_x \cos(\theta/2)^2 + g_{xp} \sin(\theta/2)^2) \sin(\theta/2)] \\ \sin(\theta/2)[2\delta + \rho^{1/3} k(g_x \theta + (g_{xp} - g_x) \sin\theta)] \end{pmatrix}. \tag{13}$$

It can be seen that when the correction functions g_x, g_{xp}, and g_δ are equal to 1, Eq. (13) reduces to the constant-σ_z model in Ref. [20], which does not account for bunch length variation.

To summarize, by modifying the CSR-kick model to account for bunch length variation within each dipole, we can calculate the CSR kick [Eq. (13)] for individual dipoles in an achromatic compressor. Specifically, for a DBA compressor, this process is particularly straightforward (see Sec. IV), as the parameters a and b for each dipole are determined (with $a = b = 0$ for the first dipole, and $a = 3$, $b = -3$ for the second dipole). For compressors with more than two dipoles, the difficulty of the analysis will depend on the specific arrangement of dipoles within the compressor.

III. SIMULATION VALIDATION

In the above sections, the modified CSR-kick model [Eqs. (6) and (13)] has been obtained for a single dipole. To validate the obtained results, in this section, we will verify the linear relation [Eq. (6)] between $(x_{\rm csr}, x'_{\rm csr}, \delta_{\rm csr})$ and $(g_x\rho^{4/3}(\theta - \sin\theta), g_{xp}\rho^{1/3}(1 - \cos\theta), g_\delta\rho^{1/3}\theta)$ with the proportionality constant k.

To verify the accuracy of the modified model, the particle tracking in the presence of the CSR effects is implemented with the ELEGANT program [33,34] to simulate the changes of horizontal coordinate deviation $(x_{\rm csr}, x'_{\rm csr})$ and energy deviation $\delta_{\rm csr}$ after passing through one dipole under different bending radius and energy chirps. To better align the simulation with theoretical assumptions, we only consider steady-state CSR. The specific parameters are listed in Table I, where the bending angle is set to 5°, and different values of ρ are obtained by varying the dipole length. One could also fix the dipole length and vary the bending angle, which would not affect the results.

As shown in Fig. 4, when considering a constant bunch length in the dipole with $h = 0$, the calculated $(x_{\rm csr}, x'_{\rm csr}, \delta_{\rm csr})$ from the constant-σ_z model [20] and the modified model are consistent and match the simulation results. However, for the compression scenario with an arbitrary energy chirp, e.g., $h = -90\ {\rm m}^{-1}$, a significant difference in the fitting coefficients between the two models emerges, with the modified model providing a closer match to the simulation results.

To verify the linear relation with a more intuitive comparison between $(x_{\rm csr}, x'_{\rm csr}, \delta_{\rm csr})$ and $(g_x\rho^{4/3}(\theta - \sin\theta), g_{xp}\rho^{1/3}(1 - \cos\theta), g_\delta\rho^{1/3}\theta)$ in Eq. (6), we calculated the proportionality constant k for both models under different values of h. As shown in Fig. 5, when h changes, the k obtained from the modified model remain closer to the theoretical values and are independent of the chirp. In contrast, the k obtained from the constant $-\sigma_z$ model deviate significantly from the theoretical values as the chirp increases. This result indicates that the modified model provides a more accurate description of the CSR effect when bunch length variations are taken into account.

TABLE I. The dipole parameters and beam parameters at the entrance of the dipole used for ELEGANT simulation.

Parameters	Symbols	Values	Units
Length of dipole	$L_{\rm B}$	0.4–2.5	m
Dipole bending angle	θ	5	Degree
Beam energy	E_0	1	GeV
Beam charge	Q	500	pC
Bunch length	σ_{z0}	20	μm
Energy spread	δ_0	0.01	%
Transverse normalized emittance	ε_{n0}	0.1	μm

IV. APPLICATION TO A DBA COMPRESSOR

Arc bunch compressors based on DBA have been performed over the last two decades in the design of free electron lasers (FELs), such as the FEL in the soft x-ray laser project at the MAX IV Laboratory [35], the compact energy-recovery linac-driven extreme ultraviolet FEL [36], and the multidisciplinary advanced research infrastructure for the generation and application of x-rays (MariX) FEL [37]. To suppress CSR effects in DBA compressors, CSR cancellation conditions were derived using the constant-σ_z model with numerical corrections [24]. In this section, we use the self-consistent modified CSR point-kick model to rederive the CSR cancellation conditions for the DBA compressor and compare them with the previous results in [24]. We assume that in the dipoles of the DBA, the Derbenev criterion [31] is satisfied, which asserts that 1D theories are valid only when $\sigma_x(\rho\sigma_z)^{-\frac{1}{3}} \ll 1$. Here, σ_x represents the transverse beam size in the dipoles, and σ_z denotes the bunch length within the dipoles.

A typical DBA is illustrated in Fig. 6. It consists of two dipoles, and a set of quadrupoles between them can be used to achieve achromatic conditions. For convenience, we denote the transfer matrix between the centers of the two dipoles as M_{c2c}. Under the achromatic condition, M_{c2c} can be expressed as [20]

$$M_{c2c} = \begin{pmatrix} -\csc\frac{\theta_2}{2}\sin\frac{\theta_1}{2} & 0 \\ r_{21} & -\csc\frac{\theta_1}{2}\sin\frac{\theta_2}{2} \end{pmatrix}. \tag{14}$$

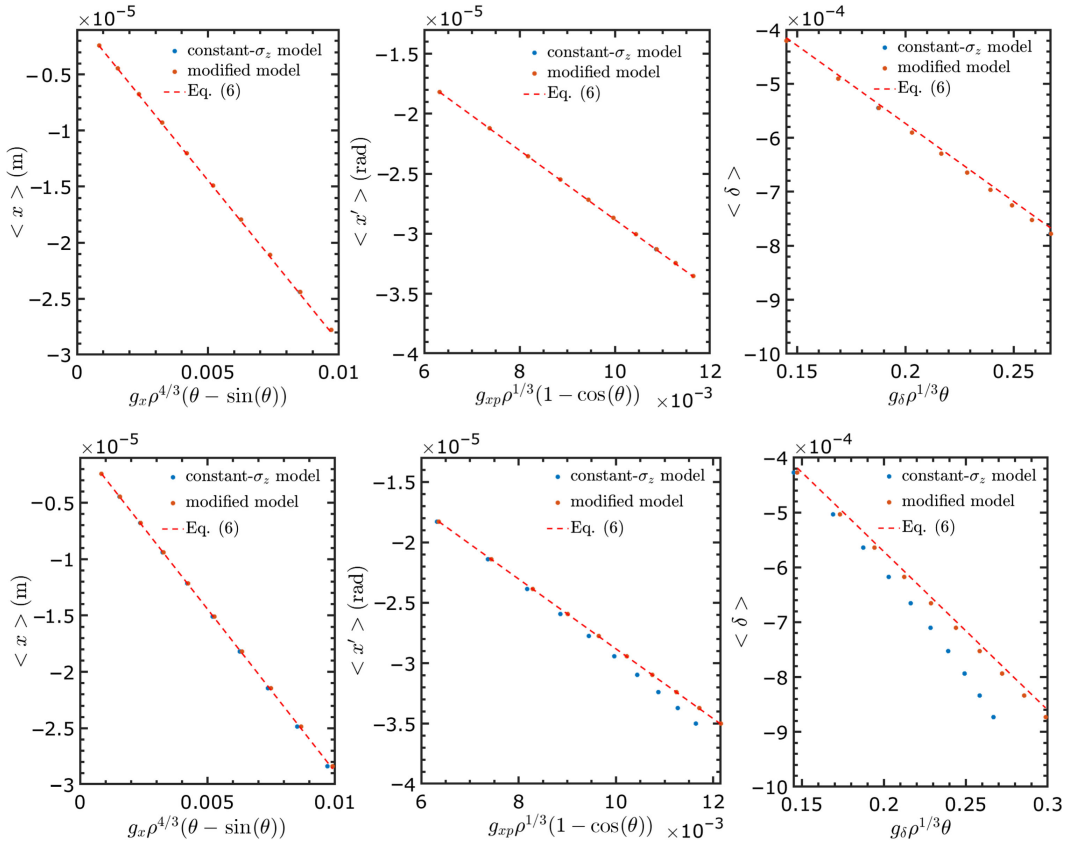


FIG. 4. Comparison of the calculated $(x_{\mathrm{csr}}, x'_{\mathrm{csr}}, \delta_{\mathrm{csr}})$ between the constant-σ_z model and modified model for $h = 0$ m^{-1} (upper) and $h = -90$ m^{-1} (lower). The blue and red points represent the simulation results processed with the constant $-\sigma_z$ model and modified models, respectively.

Here, θ_1 and θ_2 are the bending angles of the first and second dipoles, respectively.

Although the bunch length varies in a DBA compressor, the constant-σ_z model remains applicable for analysis. By assuming different constant bunch lengths in the two dipoles, one can derive the CSR-kick cancellation conditions [24]:

$$\frac{\theta_2}{\theta_1} = C^{-1/3}, \qquad r_{21} = \frac{6(1+C)\theta_2}{\rho\theta_1^2}, \tag{15}$$

where ρ is the bending radius of both dipoles.

As the compression factor C increases, the difference between this condition and the true CSR-kick cancellation conditions becomes more pronounced. To extend

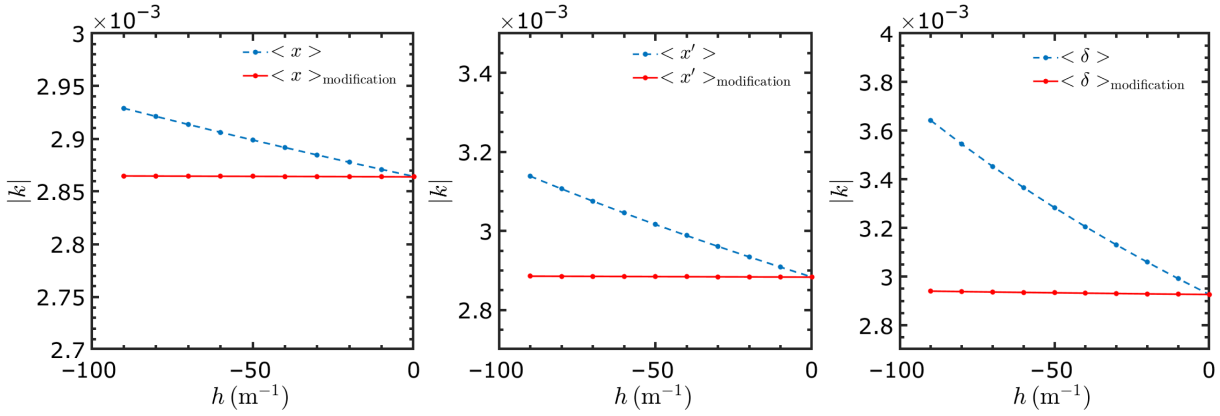


FIG. 5. The variation of the fitted proportionality constant $|k|$ with energy chirp for the constant $-\sigma_z$ model (blue curves) and modified model (red curves).

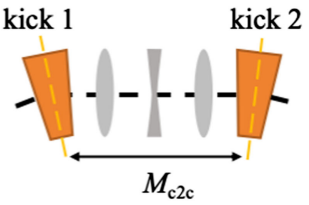


FIG. 6. Schematic of a DBA and the physical model for the CSR point-kick analysis in the DBA. The lattice section between two dipoles containing two quadrupoles and one skew quadrupole. The kicks point to the centers of the bends.

applicability to a wider range of compression factors, extensive numerical simulations and analyses were conducted, resulting in the proposed numerical corrections for C below 15 [24]:

$$\frac{\theta_2}{\theta_1} = f_1 C^{-1/3}, \qquad r_{21} = f_2 \frac{6(1+C)}{C^{1/3}\rho\theta_1}, \tag{16}$$

where f_1 and f_2 are obtained through fitting and depend only on C as

$$\begin{aligned} f_1 &= 0.142\log_{10}(C) + 1, \\ f_2 &= -0.248\log_{10}(C) + 1. \end{aligned} \tag{17}$$

Note that the compression factor C can be expressed as $C = C_1 C_2$, where C_1 and C_2 represent the bunch length compression factors of the first and second dipoles, respectively. Under the conditions of small bending angles ($\theta \ll 1$) and linear compression [Eq. (4)], C_1 and C_2 can be written as

$$C_1 = \left(1 + \frac{h_1\rho\theta_1^3}{6}\right)^{-1}, \quad C_2 = \left(1 + \frac{h_2\rho\theta_2^3}{6}\right)^{-1}. \tag{18}$$

The energy chirps in the first and second dipoles are denoted by h_1 and h_2, respectively, with

$$h_2 = C_1 h_1. \tag{19}$$

Combining Eqs. (18) and (19), we obtain

$$\frac{\theta_2}{\theta_1} = \left(\frac{C - C_1}{C_1 - 1}\right)^{1/3} C^{-1/3}. \tag{20}$$

This equation indicates that in a DBA bunch compressor, the ratio of the bending angles is linked to the compression factor, regardless of whether the CSR effect is taken into account.

In the following, we employ the modified model to rederive the CSR-kick cancellation conditions and compare them with those in Eq. (16).

Assuming that the initial transverse coordinates of the particle are $(x_0, x_0') = (0, 0)$ and the energy deviation $\delta_0 = 0$, these parameters evolve after passing through the center of the first dipole as follows:

$$\begin{aligned} X_1 &= \begin{pmatrix} x_1 \\ x_1' \end{pmatrix} = \begin{pmatrix} \rho^{4/3}k_0[g_{x1}\theta_1\cos(\theta_1/2) - 2(g_{x1}\cos(\theta_1/2)^2 + g_{xp1}\sin(\theta_1/2)^2)\sin(\theta_1/2)] \\ \sin(\theta_1/2)[\rho^{1/3}k_0(g_{x1}\theta_1 + (g_{xp1} - g_{x1})\sin\theta_1)] \end{pmatrix}, \\ \delta_1 &= g_{\delta 1}k_0\rho^{1/3}\theta_1, \end{aligned} \tag{21}$$

where k_0 is the CSR strength parameter at the entrance of the first dipole. The correction functions $g_{x1}, g_{xp1}, g_{\delta 1}$ are given by Eq. (10), with the input variable set to the bunch length variation rate for the first dipole, $1/C_1 - 1$. The transverse coordinates of this particle are then propagated through the matrix M_{c2c} to the center of the second dipole:

$$\begin{pmatrix} x_{2-} \\ x_{2-}' \end{pmatrix} = M_{c2c}X_1. \tag{22}$$

Finally, under the CSR kick in the second dipole, the coordinates transform to

$$X_2 = \begin{pmatrix} x_2 \\ x_2' \end{pmatrix} = \begin{pmatrix} x_{2-} \\ x_{2-}' \end{pmatrix} + \begin{pmatrix} \rho^{4/3}k_1[g_{x2}\theta_2\cos(\theta_2/2) - 2(g_{x2}\cos(\theta_2/2)^2 + g_{xp2}\sin(\theta_2/2)^2)\sin(\theta_2/2)] \\ \sin(\theta_2/2)[2\delta_1 + \rho^{1/3}k_1(g_{x2}\theta_2 + (g_{xp2} - g_{x2})\sin\theta_2)] \end{pmatrix}, \tag{23}$$

where $k_1 = k_0 C_1^{4/3}$ is the CSR strength parameter at the entrance of the second dipole. The correction functions $g_{x2}, g_{xp2}, g_{\delta 2}$ are defined in Eq. (11), with the input variable corresponding to the bunch length variation within the second dipole, $1/C_2 - 1$. To eliminate CSR kicks, we set $X_2 = (0, 0)^T$. Retaining only the dominant term for the bending angle yields

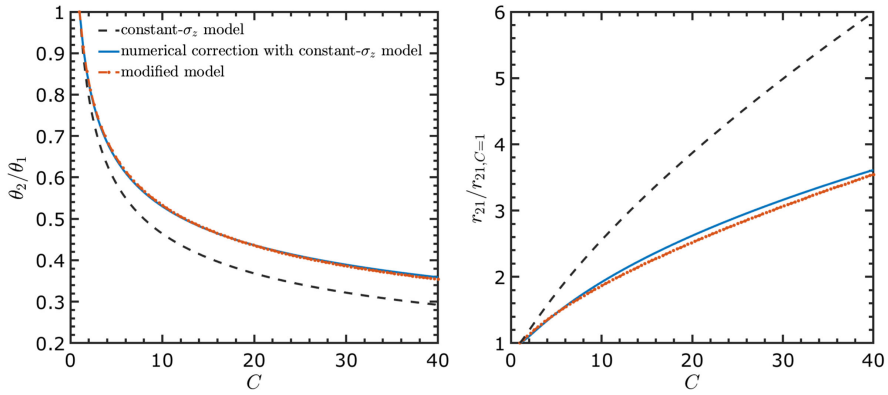


FIG. 7. The bending angle ratio θ_2/θ_1 and matrix element r_{21} as functions of the total compression ratio C.

$$\frac{\theta_2}{\theta_1} \equiv \zeta = \frac{(3g_{xp1} - 2g_{x1})^{1/4}}{(3g_{xp2} - 2g_{x2})^{1/4}} C_1^{-\frac{1}{3}},$$

$$r_{21} = \frac{6\zeta(2g_{\delta 1} - g_{xp1} + g_{xp2}\zeta C_1^{4/3})}{(3g_{xp1} - 2g_{x1})\rho\theta_1}. \tag{24}$$

Since the correction functions depend solely on the compression factors C_1 and C_2 of the dipoles in the DBA, it can be shown that the ratios θ_2/θ_1 and r_{21} depend only on the compression factors C_1, C_2, and the total compression factor C. For a given value of C, one can solve for C_1 or C_2 using Eqs. (20) and (24). This supports the conclusion in Eq. (17), which states that the numerical correction factors depend solely on C.

By incorporating Eq. (20), we calculate the corresponding θ_2/θ_1 and r_{21} for a given C using Eq. (24). We then compare these results with those from the constant $-\sigma_z$ model in Eq. (15) and the numerical corrections in Eq. (17). As shown in Fig. 7, the discrepancy between the constant $-\sigma_z$ model and the modified model is notable, emphasizing the importance of accounting for the bunch length variation within the dipole. The numerical corrections and the modified model demonstrate excellent agreement for C up to 40, with a relative discrepancy of less than 1.5% in θ_2/θ_1 and less than 4% in r_{21}, respectively. Notably, the numerical fit coefficients used in the numerical corrections were originally derived for C less than 15, which illustrates the strong generalizability of the numerical corrections from Ref. [24]. This unexpected conclusion is effortlessly obtained using the modified model, whereas previous methods may required significantly more computational resources to reach the same result.

Using the modified model, we further analyze the compression characteristics of the DBA. By applying Eqs. (20) and (24), we compute the compression factors C_1 and C_2 for the first and second dipoles, respectively, for various total compression ratio C.

The relationship between C_1 and C are shown in Fig. 8 and can be fitted using the following functional form:

$$C_1 = 1 + \frac{C-1}{2} + \lambda_1(C-1)^{\lambda_2}. \tag{25}$$

Here $\lambda_1 = -0.06841$ and $\lambda_2 = 1.197$ are obtained through fitting. Equation (25) indicates that the relationship between C_1 and C is approximately linear, rather than strictly linear. The linear term can be derived from combining Eq. (20) with the constant-σ_z CSR-kick

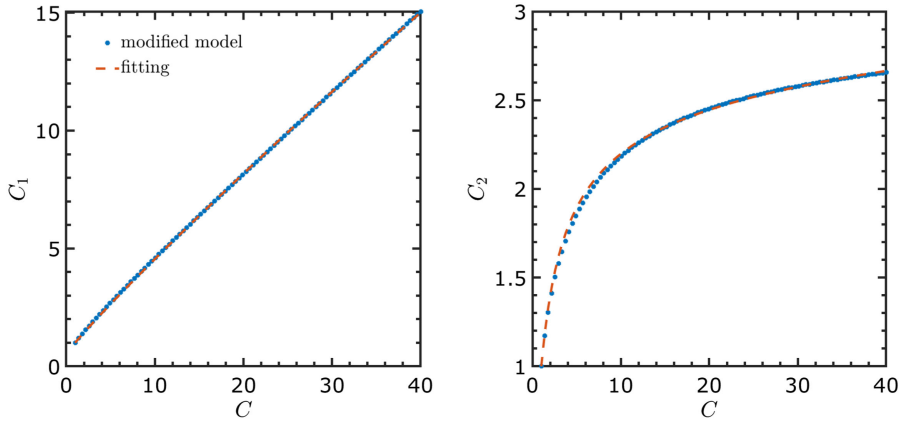


FIG. 8. Compression factors C_1 for the first dipole and C_2 for the second dipole as functions of the total compression factor C.

cancellation condition in Eq. (15), while the nonlinear term arises from corrections due to the bunch length variation.

It can be observed that the first dipole in a DBA compressor contributes the majority of the compression. For instance, C_1 and C_2 are 15 and 2.7, respecitvely, for the case where the total compression factor is $C = 40$. Furthermore, according to Eq. (25), when C_1 exceeds $(C+1)/2$, the CSR-kick cancellation condition cannot be satisfied. This result differs significantly from the conclusion of the constant-σ_z model, which requires C_1 to be exactly equal to $(C+1)/2$.

V. CONCLUSIONS AND DISCUSSION

In this paper, we have developed a modified CSR-kick model that incorporates the effects of bunch length variation by introducing simplified correction functions derived from integral expressions. These correction functions provide a more accurate description of CSR effects when the bunch length changes within dipoles. The proposed model was validated through ELEGANT simulations, confirming its accuracy.

When applied to a DBA compressor, the modified model offers a self-consistent and straightforward way to derive CSR-kick cancellation conditions, which previously required extensive parameter scans with the constant $-\sigma_z$ model. Additionally, we found that the numerical correction results remain applicable for larger compression factors C, beyond the range tested in earlier studies. Further analysis reveals that most of the bunch compression in DBA compressors is contributed by the first dipole, with an upper limit at $(C+1)/2$. Beyond this limit, the second dipole cannot effectively cancel CSR kicks due to CSR-induced coordinate shifts.

Our work neglects the impact of transverse beam parameters on the bunch length, which is reasonable for beams with small transverse size or low emittance. This simplification leads to compact correction functions. However, for larger transverse dimensions, the correction functions may become invalid, and further refinement would be needed to extend the model. In Sec. III, we considered only steady-state CSR in our simulations. To evaluate the impact of transient CSR, we also performed simulations that account for transient CSR effects. The results show that the deviation in k can reach up to 35% and is related to the drift length when entering and exiting the dipole. This indicates that steady-state CSR can only provide an approximation of the CSR effects. Future work could also consider transient CSR effects to provide a more comprehensive understanding of CSR dynamics.

The modified model can be applied to other beamline configurations, such as chicane compressors, and is equally suitable for scenarios involving bunch lengthening. Although the analytical process for more complex compressors may not be as straightforward as for DBA compressors, the model remains simple and easy to use. For instance, one could potentially simplify the correction functions through reasonable assumptions, enabling the analysis of CSR-kick cancellation conditions for chicane compressors. This could be a subject for future investigation, requiring further detailed research.

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APPENDIX: DETAILED DERIVATION OF THE MODIFIED FUNCTIONS

In a dipole, the R_{16} and R_{26} elements of the transfer matrix can be expressed as

$$R_{16}^{s\to s_f} = \rho[1-\cos(\theta-\phi)] \approx \rho\frac{(\theta-\phi)^2}{2}, \quad R_{26}^{s\to s_f} = \sin(\theta-\phi) \approx \theta-\phi. \tag{A1}$$

Here, the approximation holds for $\theta \ll 1$.

From Eqs. (1)–(5), we obtain

$$\begin{aligned} x_{\mathrm{csr}} &= \int_0^\theta k\rho^{4/3}\left(1+\frac{\xi}{1+a+b}\left[a\frac{\phi}{\theta}+b\left(\frac{\phi}{\theta}\right)^2+\left(\frac{\phi}{\theta}\right)^3\right]\right)^{-4/3}\frac{(1-\frac{\phi}{\theta})^2}{2}\theta^2 d\phi, \\ x'_{\mathrm{csr}} &= \int_0^\theta k\rho^{1/3}\left(1+\frac{\xi}{1+a+b}\left[a\frac{\phi}{\theta}+b\left(\frac{\phi}{\theta}\right)^2+\left(\frac{\phi}{\theta}\right)^3\right]\right)^{-4/3}\left(1-\frac{\phi}{\theta}\right)\theta d\phi, \\ \delta_{\mathrm{csr}} &= \int_0^\theta k\rho^{1/3}\left(1+\frac{\xi}{1+a+b}\left[a\frac{\phi}{\theta}+b\left(\frac{\phi}{\theta}\right)^2+\left(\frac{\phi}{\theta}\right)^3\right]\right)^{-4/3} d\phi, \end{aligned} \tag{A2}$$

where $\xi = h\rho\theta^3(a+b+1)/6$, and the parameters a and b are defined in Eq. (8).

According to Eq. (6), we have

$$g_x = \frac{x_{\mathrm{csr}}}{k\rho^{4/3}(\theta-\sin\theta)} \approx 3\int_0^\theta \left(1+\frac{\xi}{1+a+b}\left[a\frac{\phi}{\theta}+b\left(\frac{\phi}{\theta}\right)^2+\left(\frac{\phi}{\theta}\right)^3\right]\right)^{-4/3}\left(1-\frac{\phi}{\theta}\right)^2 d\frac{\phi}{\theta},$$

$$g_{xp} = \frac{x'_{\mathrm{csr}}}{k\rho^{1/3}(1-\cos\theta)} \approx 2\int_0^\theta \left(1+\frac{\xi}{1+a+b}\left[a\frac{\phi}{\theta}+b\left(\frac{\phi}{\theta}\right)^2+\left(\frac{\phi}{\theta}\right)^3\right]\right)^{-4/3}\left(1-\frac{\phi}{\theta}\right) d\frac{\phi}{\theta},$$

$$g_\delta = \frac{\delta_{\mathrm{csr}}}{k\rho^{1/3}\theta} \approx \int_0^\theta \left(1+\frac{\xi}{1+a+b}\left[a\frac{\phi}{\theta}+b\left(\frac{\phi}{\theta}\right)^2+\left(\frac{\phi}{\theta}\right)^3\right]\right)^{-4/3} d\frac{\phi}{\theta}. \tag{A3}$$

The approximation holds when $\theta \ll 1$. By substituting the integration variable $\tau = \phi/\theta$, we obtain the expressions for the correction functions as Eq. (7).

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